Georgios Panagopoulos | CV

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Education

Academic Qualifications

- **University of Houston** — **Houston,TX,USA**
  Ph.D. in Computer Science, Advisor : Ioannis Pavlidis — *August 2016-*
- **Harokopio University of Athens** — **Athens, Greece**
  Bachelors in Informatics and Telematics — *2010–2014*

Notable Projects

- **Masters Project (Ongoing):** *'Forecasting Driving Anomalies Using Wearable Sensors'*
- **Undergraduate Thesis:** *'Detecting Rising Stars in Dynamic Collaborative Networks'*

  I devised and implemented a novel computational methodology based on social network analysis and unsupervised machine learning to identify scientists with potential to be academic stars in the future, using their bibliographical information. The suggested methodology compiles a list of several key performance indicators (features) for each scholar in a per year basis, such as citations received, papers published, degree centrality in coauthorship network, clique's weight etc. These are monitored over time, penalized based on their temporariness and used in a clustering framework which groups the scholars into categories. The optimal number of clusters is discovered automatically using clustering validity metrics. A profile of each scholar can then be inferred through the labeling of the clusters with the used performance indicators. These labels can ultimately act as the main profile characteristics of the individuals that belong to that cluster.

Previous Employment

- **University of Houston** — **Houston**
  Teaching Assistant — *August 2017–Present*

  Software engineering.
- **NCSR Demokritos, Software Knowledge Engineering Lab** — **Athens, Greece**
  Research Intern — *May 2017–August 2017*

  I implemented a stress classification analysis for drivers based on wearable sensors and signals indicating driving behavior. I used a Hidden Markov Model trained in non-stressed sequences and then added another model trained on stressed ones. A new sequence is classified as stressed in the first case based on the loglikelihood the model gives to it, while for the second case both loglikelihoods given by each model serve as an input to classic machine learning classifiers. The results of the second approach indicated a promising accuracy level for the one of the two datasets under examination.

- **University of Houston, Computational Physiology Lab** **Houston**
  Research Assistant *August 2016–May 2017*

  I curated, combined, synchronized and preprocessed data from driving simulation experiments. The data came from 5 different sensors (wearables, thermal camera and the car) and consisted of multiple formats, for 66 subjects, 6 sessions each. experimental data coming from wearable and car sensors. I also performed exploratory data analysis to evaluate the initial hypothesis of the experiment. I performed the same for a dataset stemming from a driving experiment with a real car on a closed track.

- **University of Houston, Computational Physiology Lab** **Houston**
  Research Intern *February–March 2016*

  I performed quality control of experimental data coming from wearable signals with data visualization techniques and statistical hypothesis tests.

- **NCSR Demokritos,Software Knowledge Engineering Lab** **Athens, Greece**
  Research Assistant *September 2014–July 2016*

  During the first year, I developed a set of text mining web services using Java and Weka, that calculate computational creativity metrics from essays in three different languages, using natural language processing, unsupervised machine learning and web crawling. I also co-developed a matrix factorization technique that forms a creativity profile of a user based on the relevant metrics of her texts, with MATLAB, and a profiling server based on MySQL and Java that retrieves text and derives creativity profiles for its users. Moreover, I implemented a Java API to connect the aforementioned utilities with android serious games that are utilized in private schools in Greece, employing the Facebook & Twitter API for social media engagement. During the second year my interests swifted on Brain Computer Interfaces. I co-designed and ran workshops for university students, in Demokritos Summer School 2015 and 2016 using MATLAB, Python, Jupyter Notebooks, C++ and the Emotiv Epoc+, during which the students took part in a toy BCI experiment and were taught how to perform a rudimentary analysis on this data.In addition, I presented an interactive program to familiarize school children of age 9-15 with Brain Computer Interfaces. Overall, 121 schools attended in the span of 7 months, with over 2500 children taking part in it.Finally, I presented 'MindPong', a pong game based on Emotiv Epoc+ brain computer interface in Athens Science Festival 2016.

- **NCSR Demokritos,Software Knowledge Engineering Lab** **Athens, Greece**
  Software Engineering Intern *June 2013–September 2013*

  I was assigned to update a Web Crawler based on Delphi and set up a MySQL Database with indexes to store the crawled information in an efficient manner.

Research

Journals

- G. Panagopoulos, G. Tsatsaronis, and I. Varlamis. *"Detecting rising stars in dynamic collaborative networks"* **Journal of Informetrics** 11.1 (2017): 198-222.

Conferences

- G. Panagopoulos. *"Multi-Task Learning for Commercial Brain Computer Interfaces"*, in Proc. of the IEEE BioInformatics and BioEngineering (**BIBE**), Washington DC, USA, USA October 23-25, 2017 **[SUBMITTED]**
- P. Karampiperis, A. Koukourikos, G. Panagopoulos. *"From Computational Creativity Metrics to the Principal Components of Human Creativity"*, in Proc. of the 9th International Conference on Knowledge, Information and Creativity Support Systems (**KICSS**), Limassol, Cyprus, November 6-8, 2014
- A. Koukourikos, P. Karampiperis, G. Panagopoulos. *"Creative Stories: A Storytelling Game*

fostering Creativity", in Proc. of the 11th International Conference on Cognition and Exploratory Learning in Digital Age (**CELDA**), Porto, Portugal, October 25-27, 2014

Workshops

- G. Panagopoulos, C. Palmer, *"A Specialized Interactive Data Application for EEG Based Sleep Studies"*, Assistive Technologies for Decision making in Healthcare (ADH) in conjunction with the 10th International Conference on Pervasive Technologies Related to Assistive Environments (**PETRA**), Rhodes, Greece, June 21-23, 2017
- G. Panagopoulos, P. Karampiperis, A. Koukourikos, S. Konstantinidis, *"Creativity Profiling Server: Modelling the Principal Components of Human Creativity over Texts"*,Workshop on Deep Content Analytics Techniques for Personalized and Intelligent Services (DECAT), in conjunction with the 23rd Conference on User Modelling, Adaptation and Personalization (**UMAP**), Dublin, Ireland, June 19-July 3, 2015

Services

- Journal Reviewer: Scientometrics, Informetrics
- Publication, Publicity and Web Co-Chair : IEEE International Conference on BIoInformatics and BioEngineering, Washington DC, USA, USA October 23-25, 2017

Awards and Scholarships

- NSF Doctoral Consortium Travel Award, PETRA 2017
- Presidential Fellowship Award, University of Houston, Houston, TX, 2016
- Graduate Tuition Fellowship, University of Houston, Houston, TX, 2016
- 1st place in Open Public Data Hackathon by Greek Ministry of Administrative Reform and eGovernance on A.G.In.A.R.A. Project, Greece, 2014
- 1st place in Green Hackathon by Agro-Know on Stevia Toolkit Project, Greece, 2013

Grades

- GRE:Quantitative: 165, Verbal: 153, Analytical: 3.5
- TOEFL:106
- Masters GPA (Ongoing):3.72

Research Projects

- **University of Houston** — **Houston, Texas**
  Toyota Safety Research Project — *August 2016–Present*
  Computational Physiology Lab
- **NCSR Demokritos** — **Athens, Greece**
  Big Data Europe — *2015–2016*
  Software & Knowledge Engineering Laboratory (SKEL), Institute of Informatics & Telecommunications (IIT)
- **NCSR Demokritos** — **Athens, Greece**
  C2Learn — *2014–2015*
  Software & Knowledge Engineering Laboratory (SKEL), Institute of Informatics & Telecommunications (IIT)

Notable Coursework

Graduate Courses:

- Machine Learning, University of Houston, 2017
- Database Systems, University of Houston, 2017
- Advanced Numerical Analysis, University of Houston, 2017
- Statistical Method in Research, University of Houston, 2017

Coursera statement of accomplishment in:

- Statistical Inference, Johns Hopkins University, 2016
- Computational Neuroscience, University of Washington, 2015
- Coding the Matrix: Linear Algebra through Computer Science Applications, Brown University, 2015
- Coding the Matrix: Linear Algebra through Computer Science Applications, Brown University, 2015
- Gamification, University of Pennsylvania, 2015
- Networked Life, University of Pennsylvania, 2014
- Machine Learning, Stanford University, 2014
- Social and Economic Networks: Models and Analysis, 2014
- Data Analysis, Johns Hopkins University, 2014
- General Game Playing, Stanford University, 2013
- Web Intelligence and Big Data, Indian Institute of Technology Delhi, 2013
- Social Network Analysis, University of Michigan, 2013
- Introduction to Data Science, University of Washington, 2013

References

- Ioannis Pavlidis, Professor
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  ipavlidis@uh.edu
- Vangelis Karkaletsis, Research Director
  NCSR "Demokritos"
  vangelis@iit.demokritos.gr
- Iraklis Varlamis,Assistant Professor
  Harokopio University of Athens
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- Pythagoras Karampiperis,Research Director
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